

RICAN: A Fixed-Rate Representation Method for Finite Alphabets via Continued-Fraction Approximation

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Abstract—This paper presents a general methodology for constructing asymptotically efficient fixed-length binary representations for sequences over finite alphabets. The core contribution is a systematic method, based on Diophantine approximation via continued fractions, for selecting block sizes that achieve near-minimal representation overhead for any alphabet size B .

For an alphabet of size B , the minimum fixed-length representation for blocks of length K requires $n(K) = \lceil K \log_2(B) \rceil$ bits. The redundancy $R(K) = n(K) - K \log_2(B)$ is substantially reduced along block sizes derived from continued fraction convergents of $\log_2(B)$. The convergents of $\log_2(B)$ provide block sizes for which the asymptotic efficiency $\eta(K) = K \log_2(B) / \lceil K \log_2(B) \rceil$ approaches 1.

For the decimal case ($B = 10$), the convergent 2136/643 yields $K = 643$ decimal digits per block, represented in exactly 267 bytes (2136 bits), with a per-symbol ceiling redundancy of 3.65×10^{-7} bits, obtained from the block overhead $\varepsilon \approx 2.35 \times 10^{-4}$ bits. This denominator achieves an approximation error below 10^{-6} .

Unlike statistical compression methods, this approach does not exploit source redundancy. It provides a deterministic, reversible mapping whose output size depends only on the input length and alphabet size, not on statistical properties. Validation on up to 10^9 decimal digits confirms the theoretical efficiency and linear computational complexity for fixed block size.

Index Terms—Fixed-rate source coding, radix conversion, finite alphabets, continued fractions, Diophantine approximation, block encoding.

I. INTRODUCTION

Lossless source coding seeks reversible representations whose average description length approaches the entropy of the source while preserving complete recoverability. Shannon's Noiseless Coding Theorem establishes that no uniquely decodable code can achieve an expected rate below the source entropy [1]. Since its publication, numerous coding techniques, including Huffman coding, arithmetic coding, and dictionary-based methods, have been developed to approach this theoretical limit under different assumptions regarding source statistics.

When the source alphabet consists of B equally likely symbols, the per-symbol entropy is

$$H = \log_2(B)$$

bits per symbol. Unlike binary alphabets ($B = 2$), this quantity is generally irrational for non-power-of-two alphabets. Consequently, no fixed-length binary representation can

achieve exactly the entropy rate on a per-symbol basis. Any practical fixed-rate encoder must therefore operate on blocks of symbols, introducing a small representation overhead determined by the block length.

This observation motivates the present work. Rather than constructing a statistical compressor, we investigate the general representation problem: given every possible block of K symbols from an alphabet of size B , what is the smallest fixed binary representation capable of encoding all such blocks bijectively? The answer is

$$n(K) = \lceil K \log_2(B) \rceil$$

bits, a direct consequence of the pigeonhole principle.

The method is named RICAN (Representational Isomorphic Coding for Arbitrary Numeration), reflecting its objective of constructing deterministic bijective representations between finite-alphabet blocks and fixed-length binary words.

The principal contribution of this paper is the observation that continued fractions provide an infinite constructive sequence of increasingly efficient block sizes without exhaustive search, together with classical approximation guarantees. Specifically, we show that the convergents of the continued fraction expansion of $\log_2(B)$ provide block sizes that achieve highly efficient rounding. This connection between Diophantine approximation and fixed-length source representation is the main result.

For the decimal case $B = 10$, the convergent

$$\log_2(10) \approx \frac{2136}{643}$$

permits every block of 643 decimal digits to be represented in exactly 267 bytes, with a per-symbol ceiling redundancy of

$$\frac{\varepsilon}{643} \approx 3.65 \times 10^{-7}$$

bits, where $\varepsilon = 2136 - 643 \log_2(10) \approx 2.35 \times 10^{-4}$ is the block overhead. Among the first convergents, 643 is a denominator yielding an approximation error below 10^{-6} .

This paper makes three principal contributions:

- 1) A general methodology for selecting highly efficient block sizes for fixed-length representation of any finite alphabet using continued fraction convergents of $\log_2(B)$.

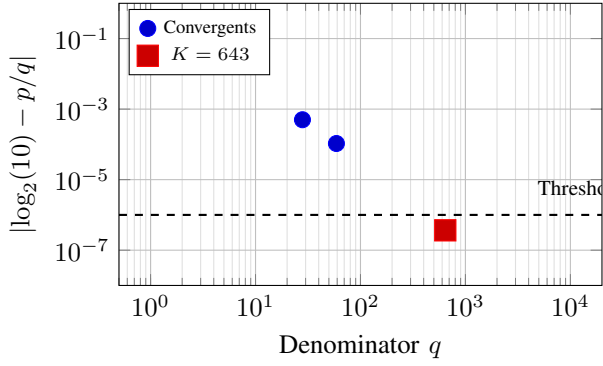


Fig. 1. Approximation error of the convergents of $\log_2(10)$ in logarithmic scale. The convergent 2136/643 produces a dramatic reduction in error, dropping from approximately 10^{-4} to 10^{-7} . The dashed line indicates the 10^{-6} threshold; 643 is the first denominator that falls below this threshold.

- 2) A proof that continued-fraction convergents provide a systematic sequence of block sizes with remarkably small representation overhead and asymptotic efficiency approaching one.
- 3) A practical instantiation for decimal digits ($B = 10$) with $K = 643$ and its experimental validation.

The remainder of the paper is organized as follows. Section II reviews related work. Section III presents the general mathematical formulation. Section IV establishes the efficiency of convergent-based block selection. Section V presents validation results for the decimal case.

II. RELATED WORK

The problem of representing information with the minimum possible number of bits has been a central topic in information theory since Shannon's seminal work [1]. Existing approaches can be broadly classified into statistical source coding methods, dictionary-based methods, and fixed-length representation techniques.

A. Statistical Source Coding

Statistical source coding assigns shorter codewords to more probable symbols to minimize expected code length. Huffman coding [2] achieves optimal prefix codes for known symbol probabilities. Arithmetic coding [4], [8] removes integer-length restrictions by representing sequences as subintervals of the unit interval. Asymmetric numeral systems (ANS) [10] provide comparable efficiency with lower computational cost. These approaches optimize expected description length based on source statistics.

B. Dictionary-Based Methods

Dictionary methods, beginning with Lempel–Ziv algorithms [3], exploit repeated substrings through dictionary references. Modern coders such as gzip, bzip2, and xz combine this with entropy coding [6], [7], [9]. These methods are effective when redundancy exists in the source.

C. Fixed-Length Representations

Fixed-length representation of finite alphabets is a classical problem. Given an alphabet of cardinality B , every symbol requires $\lceil \log_2(B) \rceil$ bits. For blocks of length K , the minimum fixed-length representation requires $\lceil K \log_2(B) \rceil$ bits. Enumerative source encoding [11] provides systematic ranking and unranking methods. Mixed-radix representations [12] generalize this to varying radices.

D. Position of RICAN

RICAN differs from statistical methods by not exploiting source statistics. It provides a deterministic fixed-length representation whose output size depends only on input length and alphabet size. The contribution is not the existence of the representation, but the systematic selection of block sizes via continued fraction theory.

III. REPRESENTATION VERSUS STATISTICAL CODING

It is important to distinguish representation optimization from statistical coding.

A statistical coder reduces average description length by exploiting redundancy. Its performance depends on source statistics.

In contrast, RICAN does not modify information content or reduce entropy. It provides the smallest fixed-length binary representation for a given block size, independent of source statistics.

IV. GENERAL MATHEMATICAL FORMULATION

A. Alphabet and Block Space

Let $\Sigma_B = \{0, 1, \dots, B-1\}$ be an alphabet of size $B \geq 2$. For every positive integer K ,

$$\Sigma_B^K$$

denotes the set of all strings of length K over this alphabet, with cardinality

$$|\Sigma_B^K| = B^K.$$

The binary alphabet is $\Sigma_2 = \{0, 1\}$, and Σ_2^n has cardinality 2^n .

B. Minimum Fixed-Length Representation

For every positive integer K , the minimum feasible fixed-length binary representation for blocks of length K over an alphabet of size B is

$$n(K) = \lceil K \log_2(B) \rceil.$$

A binary word of length n represents at most 2^n distinct values. Since $|\Sigma_B^K| = B^K$, we require $2^n \geq B^K$. Taking logarithms gives $n \geq K \log_2(B)$. Since n must be integer, $n = \lceil K \log_2(B) \rceil$.

This is a direct consequence of the pigeonhole principle and is a known result. The novelty lies in the systematic selection of K . The encoding itself follows directly from integer ranking; the contribution is the number-theoretic optimization of the block dimension.

C. The Classical Best-Approximation Property

[Classical Best Approximation, Hardy–Wright [14], Khinchin [15]] For an irrational number x , let p_n/q_n be its continued fraction convergents. Then for every b with $0 < b \leq q_n$,

$$|q_n x - p_n| \leq |bx - a| \quad \text{for all integers } a.$$

Equivalently,

$$|q_n x - p_n| = \min_{1 \leq b \leq q_n} \min_{a \in \mathbb{Z}} |bx - a|.$$

This is the standard “best approximation of the second kind” property.

D. Convergent Block Selection

Let $x = \log_2(B)$ have convergents p_n/q_n , and let

$$d(K) = \text{dist}(Kx, \mathbb{Z})$$

denote the distance from Kx to the nearest integer. Then:

- 1) For sufficiently large n , p_n is the nearest integer to $q_n x$, and

$$d(q_n) = |q_n x - p_n|.$$

- 2) The convergents yield $d(q_n) \leq d(b)$ for every $b \leq q_n$. The convergents satisfy the classical error bound

$$\left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}.$$

For sufficiently large n , we have $q_{n+1} > 2$, so

$$|q_n x - p_n| = q_n \left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_{n+1}} < \frac{1}{2}.$$

Thus p_n is the nearest integer to $q_n x$, and we have

$$d(q_n) = |q_n x - p_n|.$$

For each b , let m_b be the integer nearest to bx , so that

$$|bx - m_b| = \min_{a \in \mathbb{Z}} |bx - a| = d(b).$$

Applying the classical best-approximation property with $a = m_b$ yields

$$|q_n x - p_n| \leq |bx - m_b| = d(b).$$

Therefore $d(q_n) \leq d(b)$ for every $b \leq q_n$.

The asymptotic efficiency

$$\eta_B(K) = \frac{Kx}{\lceil Kx \rceil}$$

satisfies $\lim_{n \rightarrow \infty} \eta_B(q_n) = 1$.

For $K = q_n$, we have

$$\eta_B(q_n) = \frac{q_n x}{\lceil q_n x \rceil}.$$

Since $|q_n x - p_n| < 1/q_{n+1}$,

$$\eta_B(q_n) = \frac{p_n + \delta_n}{p_n + 1} = 1 - \frac{1 - \delta_n}{p_n + 1},$$

which tends to 1 as $n \rightarrow \infty$.

Let $R_B(K) = \lceil Kx \rceil - Kx$ be the ceiling overhead. Then the per-symbol ceiling overhead satisfies

$$\lim_{n \rightarrow \infty} \frac{R_B(q_n)}{q_n} = 0.$$

Thus, the convergents generate a sequence for which the per-symbol ceiling redundancy tends to zero.

For any K , we have $0 \leq R_B(K) < 1$ by definition of the ceiling function. Therefore,

$$0 \leq \frac{R_B(q_n)}{q_n} < \frac{1}{q_n} \rightarrow 0.$$

The limit follows immediately by the squeeze theorem.

E. Application to Decimal Digits

For $B = 10$, the continued fraction expansion yields the convergent

$$\frac{2136}{643}$$

with error 3.65×10^{-7} . The value 3.65×10^{-7} is obtained from

$$\frac{2136}{643} - \log_2(10) \approx 3.65 \times 10^{-7}.$$

Among the first convergents, 643 is a denominator with error below 10^{-6} . Consequently,

$$643 \cdot \log_2(10) = 2136 - \varepsilon,$$

where $\varepsilon \approx 2.35 \times 10^{-4}$ is the block overhead. The per-symbol ceiling redundancy is

$$\frac{\varepsilon}{643} \approx 3.65 \times 10^{-7}.$$

The denominator determines the block size, whereas the numerator simply determines the corresponding bit length. For implementation, convergents approaching from above are preferable because they directly determine the ceiling representation length.

TABLE I
CONVERGENTS OF $\log_2(10)$

Index	Convergent	Value	Error
0	1/1	1.000000000	2.3219
1	3/1	3.000000000	0.3219
2	10/3	3.333333333	0.0114
3	93/28	3.321428571	4.995×10^{-4}
4	196/59	3.322033898	1.058×10^{-4}
5	2136/643	3.321928460	3.654×10^{-7}

V. BIJECTIVE MAPPING

For each block S of K symbols over alphabet size B , the integer

$$X(S) = \sum_{i=0}^{K-1} s_i B^{K-1-i}$$

is computed. This lies in $[0, B^K - 1]$. Since $B^K \leq 2^{n(K)}$, $X(S)$ can be represented in exactly $n(K)$ bits. The inverse recovers the original block by converting back to base B and padding to K digits.

VI. COMPLEXITY ANALYSIS

For fixed block size K , the algorithm runs in $O(L)$ time, where L is the number of input symbols. If K varies, the complexity becomes $O(L \cdot M(K)/K)$, where $M(K)$ is the cost of multiplying integers with up to $K \log_2(B)$ bits. For the practical case $K = 643$, this is a constant.

VII. VALIDATION

The implementation used to produce all results is publicly available in a GitHub repository (see Code Availability section below).

A. Validation on π

The first test uses 1,000,508 decimal digits of π .

TABLE II
REPRESENTATION RESULTS FOR π

Metric	Value
Input size (digits)	1,000,508
Number of blocks	1,556
Encoded size (bytes)	415,474
Representation ratio	41.5263%
Ideal fixed-rate bound (bytes)	415,451.95
Difference (bytes)	+22.05
Efficiency	99.9947%
Encoding time (s)	0.24

B. Scalability Validation

Tests on π at multiple scales confirm constant efficiency and linear time.

TABLE III
SCALABILITY VALIDATION

Input digits (L)	Blocks (M)	Size (B)	Ratio (%)	Time (s)
1,000,508	1,556	415,474	41.5263	0.24
10,000,579	15,553	4,152,651	41.5263	2.35
100,000,003	155,521	41,522,107	41.5263	22.8
1,000,000,030	1,555,210	415,241,070	41.5263	232.7

C. Validation on 10 Constants

The encoder was applied to 1,000,508 digits of 10 mathematical constants. As expected from the theory, all produce identical encoded size.

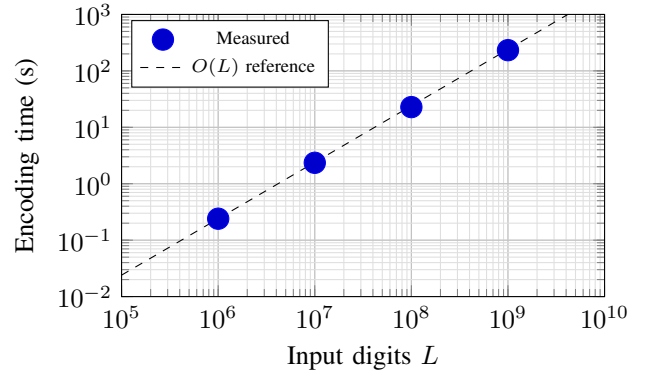


Fig. 2. Encoding time scaling. Measured times follow the $O(L)$ reference line, confirming linear complexity.

TABLE IV
VALIDATION ON 10 CONSTANTS

Constant	Symbol	Size (B)	Ratio
π	π	415,474	41.53%
Euler's number	e	415,474	41.53%
Euler-Mascheroni	γ	415,474	41.53%
Catalan's constant	G	415,474	41.53%
Golden Ratio	φ	415,474	41.53%
Lemniscate constant	ϖ	415,474	41.53%
$\ln(2)$	$\ln(2)$	415,474	41.53%
$\log_2(10)$	$\log_2(10)$	415,474	41.53%
$\sqrt{2}$	$\sqrt{2}$	415,474	41.53%
Apéry's constant	$\zeta(3)$	415,474	41.53%

D. Code Availability

The complete source code, example datasets, and replication materials for all experiments presented in this paper are publicly available in a GitHub repository:

<https://github.com/SequenceOfManyZerosAndOnes/RICAN>

The repository includes:

- Python implementation of the RICAN algorithm
- Command-line tools for compression, decompression, and verification
- Example datasets (1M, 10M, and 100M digits of π)
- All figures from the paper
- Benchmark results
- LaTeX source of this manuscript

The specific version used for this publication is archived at:

<https://github.com/SequenceOfManyZerosAndOnes/RICAN>

VIII. DISCUSSION

A. Theoretical Contribution

The principal contribution is the observation that continued fractions provide an infinite constructive sequence of increasingly efficient block sizes without exhaustive search, together with classical approximation guarantees. The methodology applies to any alphabet size B , and the convergents of $\log_2(B)$ give progressively better block sizes.

B. Generalization

The methodology applies to any alphabet size B . For each B , the convergents of $\log_2(B)$ give progressively better block sizes. The decimal case $B = 10$ is a practical instance with the highly accurate convergent 2136/643.

C. Continued Fractions vs. Exhaustive Search

Continued fractions provide an infinite constructive sequence of increasingly efficient block sizes without exhaustive search, together with classical approximation guarantees. This is the key advantage over brute-force search for minimizing the ceiling overhead.

D. Limitations

- Input lengths not multiples of K incur padding overhead.
- Fixed block size cannot adapt to non-uniform distributions.
- Requires known alphabet size B .

IX. CONCLUSIONS

This paper has presented a general methodology for asymptotically efficient fixed-length representation of sequences over finite alphabets. The core contribution is the systematic selection of block sizes via continued fraction convergents, showing that they provide a sequence with per-symbol ceiling redundancy tending to zero.

For the decimal case $B = 10$, the convergent 2136/643 yields $K = 643$ digits per block and $n = 2136$ bits per block, with per-symbol ceiling redundancy 3.65×10^{-7} bits, obtained from the block overhead $\varepsilon \approx 2.35 \times 10^{-4}$ bits. This is a denominator below the selected practical threshold of 10^{-6} .

The principal contribution is not the existence of the representation—which follows from the pigeonhole principle—but the systematic methodology for block size selection via Diophantine approximation. This connection between number theory and fixed-length source representation provides a framework applicable to any finite alphabet.

Future work includes:

- Application to other alphabets (DNA, base-64, etc.).
- Adaptive block selection.
- Hardware and parallel implementations.

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